Predicting Port Congestion and Shipment Delays at JNPT

Aiqon AI Internship take-home | Technical report | code: README.md in repository

1. Problem, data and what "congestion" means here

The dataset is **real** berth-call data from JNPT (Nhava Sheva): 464 vessel calls across four terminals (BMCT, NSICT, NSFT, NSIGT), 28 Sep - 17 Dec 2025, with arrival (berthed) / sailing times, vessel particulars and TEU handled. Cleaning (all counted in *outputs/metrics.json*): dropped 4 zero-cargo calls, 2 calls whose operation-end precedes operation-start (negative handling hours), 5 implausible turnarounds (<6 h or >45 h) → **453 calls**. TEU capacity is missing for 64 calls (median-imputed inside the model pipeline, train data only); turnaround is recomputed from timestamps rather than trusted from the raw column.

Two properties of the data drive the design. (i) *Arrival_DT is the berthing time*, so there is no anchorage queue or waiting time to measure directly. (ii) Turnaround (TAT = berth arrival to sailing) is strongly workload-driven (corr. with TEU handled = 0.59). A single "TAT is long" label would therefore mostly re-discover "big exchanges take long", which is not congestion. I define two complementary, explicitly stated targets:

Target	Ground-truth definition	Operational meaning	Base rate
Long-stay risk (primary)	TAT > 24.2 h (75th percentile of the training period)	Call overruns the normal berth window and blocks the berth for the next vessel	28% (test)
Excess-delay risk (congestion diagnostic)	TAT exceeds workload-expected TAT by > 4 h. Expected TAT = Huber regression of TAT on log TEU, log TEU^2 and vessel length, fitted on training period only	Slower than the workload justifies: queueing, equipment, labour or yard inefficiency	20% (test)

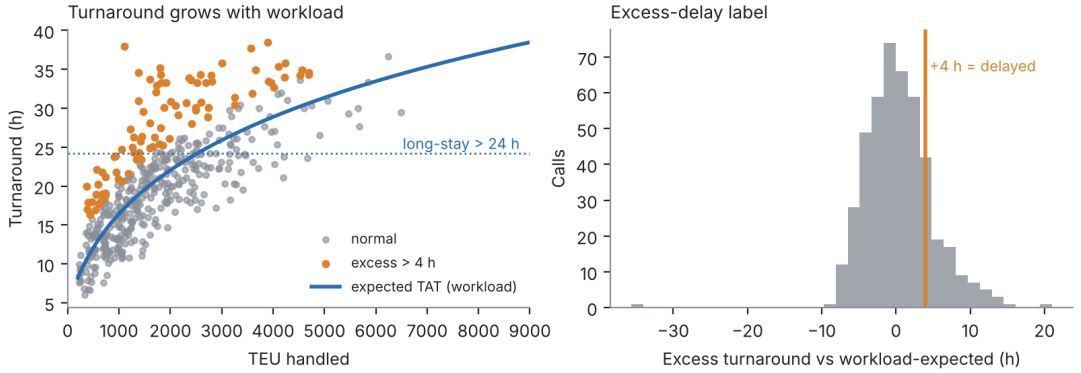


Fig. 1 - Left: TAT vs workload with the fitted expected-TAT curve; orange calls are >4 h slower than their workload justifies. Right: distribution of excess turnaround.

2. Features (all known at the moment the vessel berths - no leakage)

Every feature for a call uses only strictly earlier information: arrivals before it, departures that had already happened, and its own plan. A unit test (*tests/test_pipeline.py*) asserts that same-time and future calls never enter the state features.

Group	Features	Why it should matter
Workload / vessel	planned TEU (log), vessel length, TEU capacity, size class, import share	Moves-per-call and crane time; cargo mix drives yard handling
Berth pressure	vessels at berth (port-wide, same terminal); arrivals in last 24 h (port, terminal)	Arrival density vs berth capacity - the classic congestion signal
Historical delay	mean excess-TAT of calls sailed in last 72 h (terminal, port)	Persistence: slow terminals stay slow (temporal dynamics)
Temporal / terminal	weekend, night arrival, terminal ID	Labour rosters, terminal-specific productivity (cross-terminal variability)

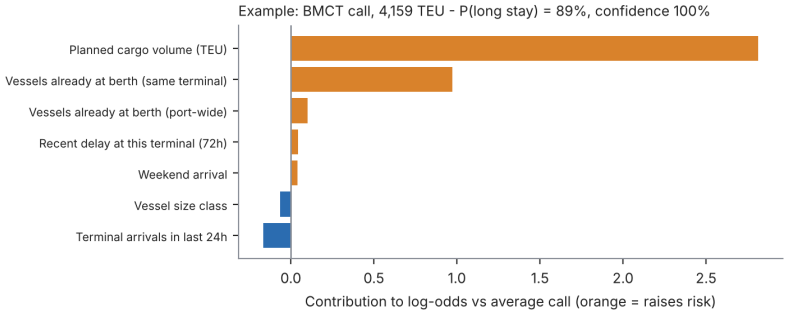
Not available in this dataset and therefore not modelled: weather, anchorage queue length, crane/berth counts, scheduled ETA. Interactions such as weather x traffic cannot be tested here; the design (state features + per-call contributions) would accept them directly.

3. Model, confidence score and explanations

Model. L2-regularised logistic regression (C=0.3; median imputation + standardisation + one-hot terminals), one per target. With ~450 calls a linear model is the right capacity and gave the best or equal out-of-time performance versus gradient boosting and random forest (Fig. 2), while making explanations exact. **Probability** of delay is the model output. **Confidence score:** 200 bootstrap re-fits; confidence = share of re-fits that agree with the final HIGH / not-HIGH decision (50-100%), reported with an 80% probability range. The HIGH cut-off is the F1-optimal threshold on out-of-fold, time-ordered predictions inside the training period, capped so at most 35% of calls are flagged (a planner's capacity assumption). Thresholds: long-stay 0.33, excess 0.24.

Explanation. For a linear model the contribution of each feature to the log-odds relative to the average call is exactly coef x (value - mean); one-hot terms are summed per feature. The top risk-raising and risk-lowering drivers are rendered as text with the actual values and a suggested operational action (e.g. *"Terminal already running slow: pre-alert yard/crane planning"*). These are

associations for decision support, not causal claims.



4. Evaluation protocol and results

Protocol. Time-ordered only (random splits would leak port state): train on the first 70% (to 2025-11-22, n=317), test on the last 30% (n=136). The expected-TAT baseline, the long-stay threshold and the HIGH cut-off are all fitted on train only. Because the test set is small, I also report pooled expanding-window CV (6 folds, everything re-fitted per fold) and bootstrap 95% CIs.

Held-out test	AUC (95% CI)	PR-AUC (base)	Flagged	Precision / recall	Lift	Brier (prior)
Long-stay	**0.89** (0.82-0.94)	0.75 (0.28)	39%	0.58 / 0.82	2.1x	0.124 (0.202)
Excess-delay	0.58 (0.47-0.68)	0.24 (0.20)	27%	0.19 / 0.26	1.0x	0.167 (0.159)

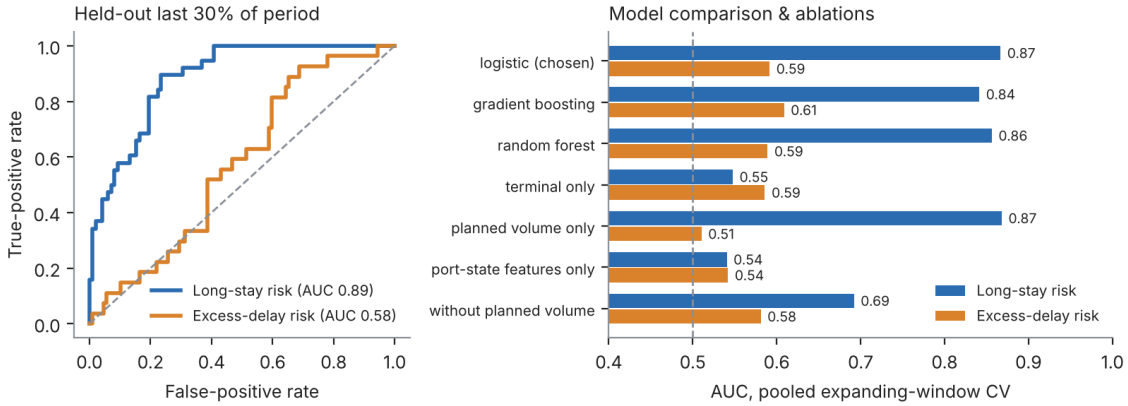


Fig. 2 - Left: ROC on the held-out period. Right: pooled out-of-time AUC for the chosen model, challengers and feature ablations.

Long-stay risk is predictable (AUC 0.89; flagging 39% of calls catches 82% of long stays at 58% precision, 2.1x the base rate; Brier 0.124 vs 0.202 for the base rate; calibration in Fig. 4 of the repo outputs). The confidence score is informative: the 68% of test calls with confidence ≥90% had 88% correct HIGH/not-HIGH decisions versus 58% for the rest.

Honest finding 1 - volume dominates. Planned TEU alone reaches CV AUC 0.87, equal to the full model (0.87); port-state features alone reach only 0.54, and removing volume drops the model to 0.69. In this data, long stays are explained mainly by the size of the exchange; berth-occupancy and arrival-density features add little (same-terminal arrivals in the last 24 h is the next-largest permutation importance, about 20x smaller than volume).

Honest finding 2 - workload-adjusted congestion is not reliably predictable from these fields. The excess-delay model has AUC 0.58 (CI 0.47-0.68 includes 0.5), lift 1.0x and a Brier score no better than the base rate; the pooled CV AUC is 0.59. The only consistent signals are terminal and vessel class (e.g. NSICT averages +1.8 h vs expectation, NSFT -1.2 h). The system therefore labels this output "low-skill: indicative only" instead of presenting a confident-looking number. I read this as a data limitation (no queue, weather or equipment data), not as proof that such delays are random.

5. From prediction to operational decisions

Signal	Decision it supports
Long-stay HIGH (≥33% prob., confidence ≥90%)	Reserve a longer berth window; do not schedule a tight follow-on vessel behind it; alert rail/road/yard teams that release may fall beyond 24 h.
Expected TAT vs 24 h window (shown in app)	Quote a realistic sailing ETA to shipper/line; plan pilot and tug slots.
Driver text: terminal busy / recent terminal delay	Stagger arrivals, or steer a flexible call to a less-loaded terminal.
Low confidence (<70%) or wide 80% range	Treat as 'watch', re-score when the ETA firms up rather than acting.

Assumptions & limitations. (1) Planned TEU is assumed known at arrival from the stow/booking plan; if only vessel particulars were known, performance falls to AUC ~0.69. (2) One port, 10 weeks, 453 calls: no seasonality can be learned (a Sep-Dec window), no cross-port transfer was tested, and test-set CIs are wide. (3) Occupancy uses calls in the table; calls missing from it

understate true berth pressure. (4) Labels are thresholds on a noisy outcome; thresholds (75th pct, +4 h) are choices, stated and configurable in *src/config.py*. (5) Explanations are associational. **Next steps:** add AIS-derived anchorage queue and scheduled-vs-actual ETA (this would give a direct waiting-time label), weather/tide, and multi-port hierarchical models; re-evaluate with rolling-origin backtests over a full year.